

Zaragoza		Equivariant Systems Theory and Observer Design for Autonomous Systems			
		Robert Mahony, Jochen Trumpf and Tarek Hamel			
M17	22/06/2026	23/06/2026	24/06/2026	25/06/2026	26/06/2026
Time	Monday	Tuesday	Wednesday	Thursday	Friday
9:00-9:30		B1: THEORY [RM]	C1: THEORY [JT]	D1: THEORY [JT]	E1: THEORY [RM]
9:30-10:00		Foundations	Lie Theory and Symmetry	Observer design	Equivariant Systems Theory
10:00-10:30		Symmetry and state		Equivariant Filter	
10:30-11:00		break	break	break	break
11:00-11:30		B2: CASE STUDY [TH]	C2: OPT. CASE STUDY [TH]	D2: CASE STUDY [RM]	E2: CASE STUDY [TH]
11:30-12:00		Attitude on SO(3)	Homography on SL(3)	Inertial Navigation	Visual Odometry
12:00-12:30	Registration			System	
12:30-13:00	A1: INTRO [RM]				
13:00-13:30	A2: THEORY [JT]	LUNCH	LUNCH	LUNCH	LUNCH
13:30-14:00	Matrix Calculus				
14:00-14:30	Matrix ODE	B3: THEORY [TH]	C3: OPT. PRACTICAL [TH]	D3: PRACTICAL [JT/TH]	E3: OPT. PRACTICAL [TH/JT]
14:30-15:00	break	Numerical implementation,	observer design	INS filter	Visual Odometry
15:00-15:30	A3: PRACTICAL [TH]	bias, time delays	for homography		
15:30-16:00	Matrix Calculus	break	break	break	
16:00-16:30	and examples	B4: PRACTICAL [JT]	C4: THEORY [RM]	D4: PRACTICAL [JT/TH]	
16:30-17:00		Observer design SO(3)	Kinematics, lifts,	INS filter	
17:00-17:30	PYTHON [JT]		velocity extensions		
17:30-18:00	set up computing				